

Conserved-reactivity control of encounter-time modality: weak-reaction theory and continuum-kernel designs

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Internal status—not a submission claim. We study reaction-time modality under one conserved centre-space reactivity integral. Within each fixed comparison, only nonnegative catalyst weights vary; transport, contact geometry, initial law, and patch supports remain fixed. For every fixed finite integer $d \geq 2$, each fixed finite m , and target times satisfying a monotone-path and contact-interior condition, we construct a d - and m -dependent longitudinal-slab family in the exact encounter quotient. Here ϵ jointly scales the OU noise, initial variances, and slab width. After first fixing a sufficiently small ϵ and then taking a sufficiently small positive installed budget B , the Doi density has at least m nondegenerate maxima; extra extrema and an absolute event-mass floor are not controlled. A weighted-space compact-positive-time mixed-jet estimate and explicit contraction/rank conditions give the corresponding weak-reaction transfer. Separately, a result-informed physical- $d = 2$ $B = 0$ free-exposure calculation with four compact slabs has a nondegenerate cusp at $t_c = 13.3280319895$ and passes declared relative-prominence shape gates with three retained maxima. The exact physical- $d = 3$ sphere kernel, evaluated in a separate result-informed chain, retains the same five-root topology and passes the same relative-shape gates. A post-G1c, result-informed calculation on one bounded reflected 207 025-state finite-volume box at $B = 0.6$ confirms a nondegenerate fold at $t_f = 10.5022583145$; its numerical choices were frozen before execution, but it was not a preregistered discovery. Neither result is an interval-global or continuum positive- B three-mode confirmation. In the broad physical- $d = 2$ family, the allocation selected from the $B \downarrow 0$ calculation was then tested without refitting at $B = 0.01$. Odd meshes $N = 113, 129$ in one fixed reflected box retain five alternating stationary roots on the saved root screen $t \leq 35$; valley-partitioned basin masses and tail checks extend through $t = 100$, and the smallest basin mass is 0.00521143. This does not exclude additional extrema after $t = 35$. It is a fixed-control, same-solver point, not an interval-global result, allocation cusp, or continuum limit. Submission remains gated by the same-family finite- B cusp and folds, parity/alignment/box continuation, and an independent killed-process calculation; positive- B physical- $d = 3$ evidence is also required for a two-/three-dimensional headline.

I. QUESTION, EVIDENCE LEVELS, AND NON-OVERLAP

First-passage and reaction-time densities can be multimodal when distinct trajectory families reach a reactive set on separated time scales [1, 2]. Strategically placed absorbing traps already produce multimodal continuum capture densities in two dimensions [3], and two-mobile-particle encounter densities have been resolved in physical dimensions two and three, including two-hump cases [4]. Exact lattice and network studies, including work by Giuggioli and co-workers, provide further multipeak mechanisms driven by transport structure or disorder [5–8].

Within each fixed control family, the unreactive transport, geometry, contact scale, initial law, and patch supports are held fixed. The sole control is a static spatial Doi reactivity field whose installed amount is conserved. Spectral survival theory for spatially distributed volumetric traps already resolves how trap arrangement changes reaction rates [9]. General heterogeneous-Robin

reaction-time theory already exists [10], as does fixed-total- reactivity optimization for steady flux [11] and encounter-history or local-time reaction theory [12–14]. We therefore claim none of those individual ingredients as new. We target their intersection with timing-modality control and model-specific bifurcation geometry. A targeted primary-source search did not locate this complete chain; that search is not proof of absence.

We use one evidence ontology: [**proved**], [**numerically verified—reduced**], [**numerically verified—discrete diagnostic**], [**numerically confirmed—exact free-exposure kernel**], [**conjectural**], [**implementation smoke pass**], [**mathematical no-go**], and [**project gate failed**]. A software or reproducibility pass is never relabeled as a scientific pass.

This internal project follows two related working manuscripts in the same research program. One develops finite encounter-reaction reductions and reduced-clock screening; the other develops finite-state localized-delivery response and catastrophe examples. Public identifiers, editor-facing copies, and an author-approved equation/code/data/figure overlap map remain release gates. Neither related manuscript is used as independent validation here. Consequently no novelty is claimed for Duhamel’s identity, catastrophe normal forms, generic

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fold powers, a first fold, or a first cusp. The non-overlapping target chain is

- fixed-budget multi-rate jet control
- sufficient at-least- m channel design (1)
- model-specific continuum jet realization.

The strongest existence statement below concerns any prescribed fixed finite m in a d - and m -dependent physical slab family for every fixed finite integer $d \geq 2$. It is pointwise rather than uniform in d and m , and is not a $d \rightarrow \infty$ or uniform $m \rightarrow \infty$ limit, or an assertion that one fixed spatial configuration has arbitrarily many modes. The exact $B = 0$ kernel shapes below are finite-parameter confirmations in both dimensions. One unchanged broad physical- $d = 2$ allocation also has event-mass-qualified modes at positive budget on two odd meshes in a fixed box. Allocation-cusp control, parity/alignment/box continuation, and an independent killed-process method remain separate publication gates.

II. CONTINUUM MODEL AND ONE PHYSICAL BUDGET

Let $q(z, t)$ be the unreacted two-particle configuration density in the Doi volume-reaction model [15], where $z = (x_1, x_2)$. The fixed conservative transport generator is $\mathcal{L}$. Write $r = x_1 - x_2$ and let $c = C_\eta(x_1, x_2)$ be the declared physical encounter-centre coordinate. With a fixed contact profile χ_a and fixed smooth centre patches ϕ_j , define

$$\begin{aligned} \kappa_{\mathbf{w}}(c) &= B \sum_{j=1}^J w_j \phi_j(c), & w_j &\geq 0, \\ \sum_j w_j &= 1, & \int_{\mathcal{C}} \phi_j(c) dc &= 1, \\ K_{\mathbf{w}}(z) &= \chi_a(r) \kappa_{\mathbf{w}}(c). \end{aligned} \quad (2)$$

The conserved installed-catalyst functional is

$$\mathcal{C}(\kappa_{\mathbf{w}}) = \int_{\mathcal{C}} \kappa_{\mathbf{w}}(c) dc = B. \quad (3)$$

It is not the integral of $K_{\mathbf{w}}$ over two-particle configuration space, a stationary-exposure weighting, or a number of reactive grid states. For an unnormalized basis the cost covector is $c_j = \int_{\mathcal{C}} \phi_j(c) dc$, and budget-tangent perturbations obey $\mathbf{c}^\top \mathbf{h} = 0$.

The killed generator, state, reaction-time density, and mass balance are

$$\begin{aligned} \mathcal{T}_\kappa &= \mathcal{L} - M_{K_{\mathbf{w}}}, & q(t) &= e^{t\mathcal{T}_\kappa} q_0, \\ f(t; \mathbf{w}) &= \langle K_{\mathbf{w}}, q(t) \rangle, & -\frac{d}{dt} \langle 1, q(t) \rangle &= f(t; \mathbf{w}). \end{aligned} \quad (4)$$

All supports, transport coefficients, contact radius, and the initial law are fixed while $\mathbf{w}$ varies. This is an amplitude-control problem, not a shape derivative for moving masks.

III. A REDUCED DESIGN LEMMA AND ITS ANCESTRY

A related encounter manuscript uses generalized-inverse-Gaussian (GIG) clocks [16] for reduced screening. For $p > 1$, $b > 0$, and $A_j > 0$, write

$$g_j(t) = Z_j^{-1} t^{-p} \exp\left(-\frac{A_j}{t} - bt\right), \quad t > 0. \quad (5)$$

Taking $m_j = m_0 R^{j-1}$, $A_j = bm_j^2 + pm_j$, and mixture weights inversely proportional to $g_j(m_j)$ makes every isolated peak occur at m_j . For each fixed finite m, p , and bm_0 , separation estimates give a finite R_{sep} such that $R \geq R_{\text{sep}}$ yields at least m nondegenerate maxima. This supporting reduced lemma is **[proved]**; it does not give the smallest threshold, exclude extra critical points, or establish event mass. The $p = 2.5$, $b = 0.01$, $R = 10$, $m = 2, \dots, 6$ root screen and the three-clock cusp are **[numerically verified—reduced]** diagnostics inherited from that screening program, not new physical evidence in this paper. GIG probabilities are not identified with the physical catalyst weights in Eq. (2).

IV. FIXED-BUDGET RESPONSE AND EXACT MODALITY JETS

For a fixed-support perturbation H , Duhamel's identity gives

$$\begin{aligned} D_\kappa f(t)[H] &= \langle H, e^{t\mathcal{T}_\kappa} q_0 \rangle \\ &\quad - \int_0^t \left\langle K_{\mathbf{w}}, e^{(t-s)\mathcal{T}_\kappa} M_H e^{s\mathcal{T}_\kappa} q_0 \right\rangle ds. \end{aligned} \quad (6)$$

This exact identity is shared background. The multi-rate object needed here is a projected jet map. For target observables $Y_a = \partial_t^a f(t_a; \mathbf{w})$, define $G_{aj} = \partial_{w_j} Y_a$ and choose a local control metric $M \succ 0$. For a general cost covector $\mathbf{c}$,

$$\tilde{G} = G - \frac{(GM^{-1}\mathbf{c})\mathbf{c}^\top}{\mathbf{c}^\top M^{-1}\mathbf{c}}. \quad (7)$$

If the control lies in the simplex interior and $\text{rank } \tilde{G} = q$, the unique budget-preserving perturbation of minimum M -norm that realizes a target infinitesimal jet $\mathbf{y}$ is

$$\mathbf{h}_* = M^{-1} \tilde{G}^\top (\tilde{G} M^{-1} \tilde{G}^\top)^{-1} \mathbf{y}. \quad (8)$$

Equations (7)–(8) are **[proved]** constrained linear algebra. They are local, not a global finite-amplitude control theorem. A physical controllability claim still requires a continuum-stable positive lower bound on $\sigma_{\min}(\tilde{G} M^{-1/2})$.

Writing $F = f_t$, a nondegenerate one-control fold satisfies

$$f_t = f_{tt} = 0, \quad f_{ttt} f_{t\theta} \neq 0. \quad (9)$$

Its minimal persistence jet is

$$\mathcal{J}_{\text{fold}} = \{f_t, f_{tt}, f_{ttt}, f_{t\theta}, f_{tt\theta}\}. \quad (10)$$

With two independent budget controls, a cusp satisfies

$$f_t = f_{tt} = f_{ttt} = 0, \quad f_{tttt} \neq 0, \quad (11)$$

and the projected gradients of f_t and f_{tt} have rank two. Its exact persistence jet is

$$\mathcal{J}_{\text{cusp}} = \{f_t, f_{tt}, f_{ttt}, f_{tttt}, f_{t\theta_i}, f_{tt\theta_i}, f_{ttt\theta_i} : i = 1, 2\}. \quad (12)$$

Full joint C^3 and C^4 convergence are convenient sufficient conditions for the fold and cusp, respectively. A cusp creates only one additional local max-min pair. A trimodal conclusion also needs a remote pair with uniform root, curvature, prominence, and tail margins.

V. WEAK-REACTIVITY CONTINUUM BRIDGE AND REMAINING GLOBAL TARGET

Write $K_{B,w} = BV_w$ and $A_{B,w} = \mathcal{L} - BM_{V_w}$. First let $\mathcal{Q}_{\text{box}}$ be a bounded Lipschitz quotient with reflecting boundaries and a reversible free invariant density bounded above and below. The free forward operator is then similar to a nonpositive self-adjoint weighted-Neumann operator. Bounded indicator killing preserves an analytic semigroup. For $q_0 \in L^2$ every finite time/control mixed jet exists on $[\tau, T]$, $\tau > 0$, even though the contact indicator is discontinuous. If $w(\theta) = w_* + P\theta$ and $U_i = V_{P\theta_i}$, the first two exact state sensitivities satisfy

$$\begin{aligned} \partial_t s_i &= A_{B,w} s_i - BU_i q, & s_i(0) &= 0, \\ \partial_t s_{ij} &= A_{B,w} s_{ij} - BU_i s_j - BU_j s_i, & s_{ij}(0) &= 0. \end{aligned} \quad (13)$$

The observable must also be differentiated. In particular,

$$\begin{aligned} \partial_i f_B &= B\{\langle U_i, q \rangle + \langle V_w, s_i \rangle\}, \\ \partial_{ij} f_B &= B\{\langle U_i, s_j \rangle + \langle U_j, s_i \rangle + \langle V_w, s_{ij} \rangle\}. \end{aligned} \quad (14)$$

Omitting the direct terms in Eq. (14) gives the wrong control jet. Perturbative treatments of spatially nonuniform partial absorption precede this calculation [17, 18]; the object used here is the transient, budget-projected mixed time/control jet.

The analytically tractable leading object is the normalized density and its free-exposure limit,

$$F_B(t, \theta) = B^{-1} f_B(t, \theta), \quad G(t, \theta) = \langle V_{w(\theta)}, e^{t\mathcal{L}} q_0 \rangle. \quad (15)$$

The reactivity-times-occupancy identity and its occupation-time/local-time interpretations are standard [12, 13, 19, 20]; here they provide the budget-normalized leading object rather than a new reaction formalism. Let Θ be a compact interior control set and let Θ_δ be

a complex control tube. With $v_{\infty, \delta} = \sup_{\Theta_\delta} \|V_w\|_\infty$, $v_{2, \delta} = \sup_{\Theta_\delta} \|V_w\|_2$, and κ_π the reversible-similarity condition number, Dyson expansion followed by time/control Cauchy estimates gives, for every fixed r and multi-index α ,

$$\begin{aligned} & \sup_{[\tau, T] \times \Theta} |\partial_t^r \partial_\theta^\alpha (F_B - G)| \\ & \leq r! \alpha! \left(\frac{2}{\tau}\right)^r \delta^{-|\alpha|} v_{2, \delta} \kappa_\pi \\ & \times \left[e^{(3/2)Bv_{\infty, \delta} T} - 1 \right] \|q_0\|_2 = O(B). \end{aligned} \quad (16)$$

This is a **[proved]** compact-positive-time continuum mixed-jet bridge. The first Dyson correction has an explicit $O(B^2)$ remainder on the same complex neighborhood.

a. Unbounded weighted-space corollary. For the physical cylinder $\mathcal{Q}_d^\infty = \mathbb{R}^2 \times \mathbb{T}_W^{d-1}$, let π be the Gaussian-uniform reversible invariant density and set

$$X_\pi = L^2(\pi^{-1} dx), \quad \|V\|_{X_\pi^*} = \|V\|_{L^2(\pi dx)}. \quad (17)$$

The map $u \mapsto q = \pi u$ is unitary from $L^2(\pi dx)$ to X_π . Thus, for $q_0 \in X_\pi$ and bounded fixed- ϵ killing profiles, the same proof gives Eq. (16) with $\kappa_\pi = 1$, $v_{2, \delta}$ replaced by $\sup_{\Theta_\delta} \|V_w\|_{X_\pi^*}$, and $\|q_0\|_2$ replaced by $\|q_0\|_{X_\pi}$. This is the version used by the unbounded-cylinder theorem below; it is not obtained from a bounded-box similarity constant.

b. Quantitative persistence criterion. With the continuous extension $F_0 = G$, let $H_B = (F_{B,t}, F_{B,tt})$ for a fold, with variables $x = (t, \theta)$, or let $H_B = (F_{B,t}, F_{B,tt}, F_{B,ttt})$ for a cusp, with variables $x = (t, \theta_1, \theta_2)$. Suppose $H_0(x_0) = 0$, $A_0 = DH_0(x_0)$ is invertible, and the closed ball $\overline{B_r(x_0)}$ lies inside positive time and the simplex interior. If

$$\begin{aligned} & \sup_{x \in \overline{B_r(x_0)}} \|I - A_0^{-1} DH_B(x)\|_2 \leq \frac{1}{2}, \\ & \|A_0^{-1} H_B(x_0)\|_2 \leq \frac{r}{2}, \end{aligned} \quad (18)$$

then $x \mapsto x - A_0^{-1} H_B(x)$ is a contraction of the ball into itself. It has one zero x_B there,

$$\|x_B - x_0\|_2 \leq 2\|A_0^{-1} H_B(x_0)\|_2, \quad (19)$$

and $DH_B(x_B)$ remains invertible. For a cusp, or any unfolding with at least two simplex-tangent controls, nondimensionalize time once and use one frozen dimensionless tangent metric. In that metric define the two-row projected control response

$$R_B(x) = \begin{pmatrix} \nabla_\theta (F_B)_t(x) \\ \nabla_\theta (F_B)_{tt}(x) \end{pmatrix}, \quad (20)$$

with R_0 defined by $F_0 = G$. On $U = \overline{B_r(x_0)}$, suppose

$$\begin{aligned} s_* &:= \inf_{x \in U} \sigma_2(R_0(x)) > 0, \\ \varepsilon_R(B) &:= \sup_{x \in U} \|R_B(x) - R_0(x)\|_2 < s_*. \end{aligned} \quad (21)$$

Weyl's inequality then gives $\inf_{x \in U} \sigma_2(R_B(x)) \geq s_* - \varepsilon_R(B) > 0$, including at the displaced root x_B . The finite collection of entrywise bounds from Eq. (16) controls $\varepsilon_R(B)$. The raw response for f_B is BR_B : normalized rank can persist while its absolute smallest singular value and the event rate vanish as $B \downarrow 0$. For a one-control fold, Eq. (18) and the resulting invertibility of $DH_B(x_B)$ supply the local transfer; no rank-two condition is imposed on its 2×1 control response. For a cusp, the contraction criterion together with Eqs. (20)–(21) and the nonzero quartic margin supplies the explicit small- B transfer conditions.

For the exact slab quotient, the free clocks factor as $g_j(t) = a_j(t)c_d(t)$. With $g = (g_1, g_2, g_3)^\top$, define

$$\mathcal{D}(t) = \begin{pmatrix} g'(t)^\top \\ g''(t)^\top \\ g'''(t)^\top \end{pmatrix}, \quad \Delta(t) = \det \mathcal{D}(t). \quad (22)$$

Analytical modality criteria for finite mixtures are classical [21]; the following Wronskian-like determinant and its conserved-simplex identity are specific to the present encounter clocks. A rank-two zero of Δ whose normalized null vector $\mathbf{w}_*$ is strictly positive is an interior free-exposure cusp provided $g'''(t_*)^\top \mathbf{w}_* \neq 0$ and the two budget-tangent rows $g'(t_*)^\top P, g''(t_*)^\top P$ have rank two. The exact identity

$$\begin{aligned} \det DC_0(t_*, 0) &= [g'''(t_*)^\top \mathbf{w}_*] \\ &\quad \times \det \begin{pmatrix} g'(t_*)^\top P \\ g''(t_*)^\top P \end{pmatrix} \\ &= \det[P, \mathbf{w}_*] \Delta'(t_*) \end{aligned} \quad (23)$$

turns spatial configuration into a computable cusp-design criterion. These standard ingredients provide a model-specific route from a quantified free-exposure singularity to a nearby weak-Doi singularity. A finite-budget, event-mass-qualified fixed-control point is given below; establishing the same-family cusp, both folds, and their converged continuation remains a separate gate.

The theorem is not uniform on the long scale $t = O(B^{-1})$, does not show that the pilot value $B = 0.6$ is small, and does not by itself exclude late extra critical points. Event mass on a fixed window is $O(B)$, so an observability lower bound must overlap the theorem's small- B upper bound. A Scharfetter–Gummel/FEM C^1 jet estimator and an independent solver remain separate gates.

A. Direct fixed-finite-mode theorem in every fixed finite $d \geq 2$

The GIG family is not needed to prove physical existence. Fix $d \in \mathbb{N}$ with $d \geq 2$, $D_0, \gamma, W, \rho, s_0, u_0 > 0$, $0 < a < W/2$, and $z_0 \neq \bar{z}$, and let the longitudinal mid-

point satisfy

$$\begin{aligned} dZ_t &= -\gamma(Z_t - \bar{z}) dt + \epsilon \sqrt{D_0} dW_t, \\ Z_0 &\sim N(z_0, \epsilon^2 s_0^2), \\ \mu(t) &= \bar{z} + (z_0 - \bar{z})e^{-\gamma t}. \end{aligned} \quad (24)$$

For the relative quotient of two independent equal-diffusivity walkers, use

$$\begin{aligned} dR_{\parallel,t} &= -\gamma R_{\parallel,t} dt + 2\epsilon \sqrt{D_0} dW_{\parallel,t}, \\ dR_{\perp,t} &= 2\epsilon \sqrt{D_0} d\mathbf{W}_{\perp,t} \pmod{W}, \\ R_{\parallel,0} &\sim N(r_{\parallel,0}, \epsilon^2 u_0^2), \\ R_{\perp,0} &\sim \text{WN}(r_{\perp,0}, \epsilon^2 \Sigma_{\perp,0}), \end{aligned} \quad (25)$$

where WN denotes a wrapped Gaussian on the transverse torus and $r_{\perp,0} \in \mathbb{T}_W^{d-1}$ and $\Sigma_{\perp,0} \in \mathbb{R}^{(d-1) \times (d-1)}$ is positive definite. The three initial variables, all quotient Brownian drivers, and the midpoint driver are mutually independent. For each fixed $\epsilon > 0$, let π_ϵ be the product invariant density: Gaussian with mean $\bar{z}$ and variance $\epsilon^2 D_0/(2\gamma)$ in Z , Gaussian with mean zero and variance $2\epsilon^2 D_0/\gamma$ in $R_{\parallel}$, and uniform on the transverse torus. Direct comparison of the declared Gaussian factors gives $s_0^2 < D_0/\gamma$ and $u_0^2 < 4D_0/\gamma$ as sufficient and necessary conditions for their contribution to $q_0 \in X_{\pi_\epsilon} = L^2(\pi_\epsilon^{-1} dx)$; the wrapped Gaussian factor is square integrable against the uniform torus density. These are the fixed- ϵ hypotheses used below, and no weighted-norm estimate is uniform as $\epsilon \downarrow 0$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$ use normalized slabs

$$\begin{aligned} \phi_{j,\epsilon}(z) &= \frac{\exp[-(z - c_j)^2/(2\epsilon^2 \rho^2)]}{\sqrt{2\pi}\epsilon\rho}, \\ \kappa &= \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\epsilon}. \end{aligned} \quad (26)$$

Every control therefore has the same full centre-space integral B . When the local killing field has units T^{-1} , the normalized centre-space profiles imply $[B] = L^d T^{-1}$ in physical dimension d ; equal numerical values of this dimensional budget are not compared across dimensions. Choose distinct target times $\tau < t_1 < \dots < t_m < T$ along the monotone deterministic midpoint path and set $c_j = \mu(t_j)$. Let I_* be the union of disjoint neighborhoods of these times and let $r_*(t) = (r_{\parallel,0} e^{-\gamma t}, r_{\perp,0})$ be the deterministic relative mean. We assume the explicit contact-interior margin $\sup_{t \in I_*} |r_*(t)|_{\text{mi}} \leq a - \eta$ for some $\eta > 0$, where $\{|r|_{\text{mi}} < a\}$ is the embedded minimum-image d -ball. Then the contact probability $c_{d,\epsilon}(t)$ and its first two time derivatives differ from $(1, 0, 0)$ by at most a polynomial in ϵ^{-1} times $\exp(-q/\epsilon^2)$.

Independence of the declared initial law and drivers makes midpoint and relative motion independent. The exact free-exposure channel therefore factorizes as

$$g_{j,\epsilon}(t) = \frac{c_{d,\epsilon}(t)}{W^{d-1} \sqrt{2\pi}\epsilon S(t)} \exp\left[-\frac{(c_j - \mu(t))^2}{2\epsilon^2 S^2(t)}\right], \quad (27)$$

where $S^2(t) = s^2(t) + \rho^2$; the physical convolved variance is $\epsilon^2 S^2(t)$. For $t = t_j + \epsilon y$ and $r = 0, 1, 2$,

$$\epsilon^{r+1} \partial_t^r g_{j,\epsilon}(t_j + \epsilon y) \longrightarrow \partial_y^r A_j(y),$$

$$A_j(y) = \frac{\exp[-\mu'(t_j)^2 y^2 / (2S^2(t_j))]}{W^{d-1} \sqrt{2\pi} S(t_j)}. \quad (28)$$

The convergence is uniform in C^2 on every fixed y interval. For $i \neq j$, monotonicity of μ instead gives the cross-channel bound

$$\sup_{|t-t_j| \leq L\epsilon} |\partial_t^r g_{i,\epsilon}(t)| \leq C_{ijr} \epsilon^{-M_r} e^{-q_{ij}/\epsilon^2}, \quad r = 0, 1, 2. \quad (29)$$

a. Theorem (at least m modes in every fixed finite dimension). Fix finite integers $d \geq 2$ and $m \geq 1$, the dimension-sized model data, the target times, and a compact interior weight set $w_j \geq w_{\min} > 0$, $\sum_j w_j = 1$. There is an $\epsilon_0 > 0$ such that for every $0 < \epsilon < \epsilon_0$ the exact continuum free-exposure mixture $G_\epsilon = \sum_j w_j g_{j,\epsilon}$ has exactly one nondegenerate maximum in each of m disjoint $O(\epsilon)$ neighborhoods of the target times and at least one local minimum between consecutive neighborhoods. For each such fixed ϵ , there is a $B_0(\epsilon) > 0$, uniform over the declared weight set, such that the full Doi density has the same m certified maxima for every $0 < B < B_0(\epsilon)$.

To prove the first statement, choose a common local interval on which every A_j'' is strictly negative. Equation (28) supplies opposite endpoint slopes of order ϵ^{-2} and negative curvature of order ϵ^{-3} ; Eq. (29) is exponentially smaller. Thus G'_ϵ is strictly decreasing and changes sign once on each interval. Its sign at consecutive interval endpoints also forces an intervening local minimum. For fixed ϵ , the catalyst profiles are bounded and the unbounded weighted-space corollary following Eq. (16) gives $f_{B,\epsilon}/B \rightarrow G'_\epsilon$ in $C^2[\tau, T]$, uniformly over the compact weight set. The strict slope and curvature margins persist for sufficiently small positive B . This proves the theorem without identifying physical weights with GIG probabilities.

Leading peak heights can be equalized by $w_j = S(t_j)/\sum_i S(t_i)$, for which the ratio of the smallest to largest certified free-exposure peak tends to one. This does not provide an absolute event-mass floor: for fixed ϵ , Doi mass on each local window is B times the free-exposure area plus $O(B^2)$. The construction is a d -dependent, m -dependent, ϵ -dependent longitudinal slab family, not one fixed configuration for arbitrary m ; it certifies at least m maxima and does not exclude additional extrema. The limits are sequential—first fix a sufficiently small ϵ , then take $B < B_0(\epsilon)$ —with no uniform interchange. No constant, dimensional budget, value of B_0 , amplitude, or event mass is uniform or compared across dimensions; no $d \rightarrow \infty$ limit is claimed. The theorem is not for arbitrary localized patches and deliberately keeps the deterministic relative path inside contact near the designed peaks.

b. Boundary of the reduced analogy. No GIG-to-Doi universality is claimed. The direct theorem removes the

need for such a realization to prove the fixed-finite- m , at-least- m statement, while realizing a selected GIG mixture with uniform mixed-jet error remains open. Known Doi–Robin small-target calibrations [22] do not supply that result, and no conclusion below depends on it.

VI. PHYSICAL $d = 2$ CONFIGURATIONS AND MODALITY TRANSITIONS

The unbounded physical slab model uses two equal-diffusivity particles on $\mathbb{R} \times \mathbb{T}_W$ with identical longitudinal Ornstein–Uhlenbeck confinement and free transverse diffusion. Define

$$\begin{aligned} z &= (X_1 + X_2)/2, \\ r_{\parallel} &= X_1 - X_2, \\ r_{\perp} &= Y_1 - Y_2 \pmod{W}. \end{aligned} \quad (30)$$

Because transport and the catalyst slabs are independent of common transverse translation, integrating that coordinate closes exactly. The physical $d = 2$ process becomes a three-dimensional PDE,

$$\begin{aligned} \partial_t q &= \frac{D}{2} \partial_{zz} q + \gamma \partial_z [(z - \bar{z})q] + 2D \partial_{r_{\parallel} r_{\parallel}} q \\ &\quad + \gamma \partial_{r_{\parallel}} [r_{\parallel} q] + 2D \partial_{r_{\perp} r_{\perp}} q - \mathbf{1}_{r_{\parallel}^2 + \delta_W (r_{\perp})^2 < a^2} \kappa_w(z) q. \end{aligned} \quad (31)$$

The condition $a < W/2$ keeps the true circular contact set away from the torus cut locus. For a slab, the normalized full centre profile is $\phi_j(c) = \phi_j^{\parallel}(z)/W$. Hence $\kappa_w(z) = (B/W) \sum_j w_j \phi_j^{\parallel}(z)$ and the one physical installed amount remains $W \int \kappa_w(z) dz = B$. This exact quotient represents longitudinal catalyst slabs, not arbitrary localized two-dimensional patches. The same symmetry gives a four-dimensional quotient with a true spherical contact set in physical $d = 3$. The exact free kernels below and the direct theorem use this unbounded OU cylinder.

The G1a–G1d finite-volume calculations instead use a distinct reflected truncation:

$$\begin{aligned} z &\in [-0.25, 1.85], & r_{\parallel} &\in [-1.8, 1.8], \\ r_{\perp} &\in [-0.5, 0.5] \pmod{1}. \end{aligned} \quad (32)$$

with zero normal flux at the first two coordinate boundaries. No box-to-unbounded limit has been established. Moreover, the G1 parameters ($D = 0.0045$, three centres 0.48, 0.67, 0.86, patch half-width 0.08) and the four-slab parameters below are separate configurations, not points on one validated continuation branch.

A. Relative-prominence-qualified four-slab $B = 0$ design

We next evaluate the exact unbounded OU and periodic free kernels, rather than a spatial grid, for a result-informed four-slab geometry. The fixed parameters are

$D = 0.002$, $\gamma = 0.1$, $\bar{z} = 0.95$, $W = 1$, and contact radius $a = 0.16$. The initial compact bump has half-width 0.004; normalized compact catalyst bumps of half-width 0.008 are centred at

$$(c_0, c_1, c_2, c_3) = (0.35, 0.60, 0.75, 0.90). \quad (33)$$

Each continuum channel is the exact product of its mid-point exposure and the true disk-contact probability, including the $1/W$ full-budget factor. The plotted mixture is $G = \lim_{B \downarrow 0} f_B/B$, not a reaction-time probability density with positive event mass. Accordingly, this subsection tests only relative $B = 0$ shape gates and makes no observability claim.

On the affine slice $w_0 = 0.28$, the equations $G_t = G_{tt} = G_{ttt} = 0$ give

$$\begin{aligned} t_c &= 13.3280319895, \\ \mathbf{w}_c &= (0.28000000, 0.23019485, 0.20932396, 0.28048119). \end{aligned} \quad (34)$$

The largest scaled residual is 1.10×10^{-12} , $t_c^4 G_{tttt}/G = -42.8118$, and the dimensionless unfolding singular-value ratio is 0.256405. Thus the cusp has nonzero quartic and rank-two control margins. A frozen scan of inward-normal steps $s = 0.02, 0.03, \dots, 0.20$ selected $s = 0.11$ by maximizing the minimum catalyst weight, not the pre-existing passing hint $s = 0.15$. The selected weights are

$$\mathbf{w}_* = (0.28000000, 0.26950477, 0.10658775, 0.34390748). \quad (35)$$

On the declared floating-point screen $t \in [0.1, 100]$ with spacing 0.002, five alternating simple sign-changing roots are retained at

$$\begin{aligned} &3.20404_{\max}, \quad 5.08547_{\min}, \quad 8.68847_{\max}, \\ &13.32803_{\min}, \quad 22.66010_{\max}. \end{aligned} \quad (36)$$

The peak-height ratio $\min(P_1, P_2, P_3)/\max(P_1, P_2, P_3)$ is 0.85413, while the two valley-to-smaller-adjacent-peak ratios are 0.66679 and 0.83754. Hence the peak-ratio gate passes its declared floor 0.10 and both valleys pass the declared ceiling 0.85 (Fig. 1).

Coarse, primary, and fine quadratures agree in cusp time within 1.43×10^{-11} and in weights within 1.05×10^{-14} ; the primary/fine scaled-quartic difference is 8.84×10^{-9} . A separate polar-disk quadrature agrees with the half-chord contact integral within 6.60×10^{-15} at four fixed times; primary/fine selected-root times agree within 7.11×10^{-15} . These checks support **[numerically confirmed—exact free-exposure kernel]**. They do not supply interval-exhaustive root isolation, a positive- B persistence radius, or an independent killed-PDE solver; the machine-readable flags `continuum_verified`, `finite_B_Doi_verified`, and `project_gate_passed` therefore remain false.

B. Physical-d=3 exact sphere-kernel analogue

We next keep the same longitudinal transport, initial and catalyst bumps, centres, contact radius, period, and

affine slice, but replace the one periodic relative coordinate by two. Contact is the true sphere in $\mathbb{R} \times \mathbb{T}_W^2$, and the full-budget slab factor is $1/W^2$. Both transverse relative coordinates start from independent compact bumps of half-width 0.004 centred at zero. The exact Fourier–Bessel sphere kernel gives

$$\begin{aligned} t_c^{(3)} &= 12.8097399605, \\ \mathbf{w}_c^{(3)} &= (0.28000000, 0.18220393, \\ &\quad 0.20766999, 0.33012608), \\ (t_c^{(3)})^4 G_{tttt}/G &= -39.8723, \\ \sigma_{\min}/\sigma_{\max} &= 0.240300. \end{aligned} \quad (37)$$

All cusp weights are interior and the unfolding has rank two. Of the 19 frozen inward steps, 11 pass every relative-shape gate; the predeclared priority selects $s = 0.10$ and

$$\mathbf{w}_*^{(3)} = (0.28000000, 0.21134977, 0.11201164, 0.39663859). \quad (38)$$

On the same $[0.1, 100]$, spacing-0.002 floating-point screen, the retained alternating roots are

$$\begin{aligned} t_1 &= 3.07471 \text{ (max)}, \quad t_2 = 5.50128 \text{ (min)}, \\ t_3 &= 8.26705 \text{ (max)}, \quad t_4 = 12.80974 \text{ (min)}, \\ t_5 &= 21.46418 \text{ (max)}. \end{aligned} \quad (39)$$

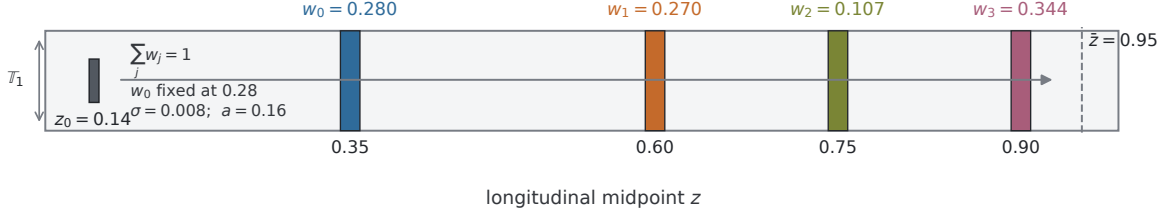
The peak-height ratio is 0.63381, and the two valley ratios are 0.76924 and 0.84480. Thus the exact sphere-kernel shape retains three relative-prominence-qualified maxima, although its second-valley margin below 0.85 is only about 0.0052 and is a genuine robustness warning (Fig. 2).

Fine-to-primary differences are 1.28×10^{-11} in cusp time, 1.50×10^{-14} in weight max norm, 4.05×10^{-10} in the scaled fourth derivative, and 6.18×10^{-13} in the largest selected root time. A direct spherical-coordinate quadrature of the pointwise product of one OU and two periodic heat kernels—which does not use the Bessel disk tensor—agrees at four fixed times within 1.08×10^{-14} . This is an independent representation of the free kernel, not an independent PDE solver. The calculation is result-informed and $B = 0$; its interval, finite- B , independent-solver, and project flags remain false.

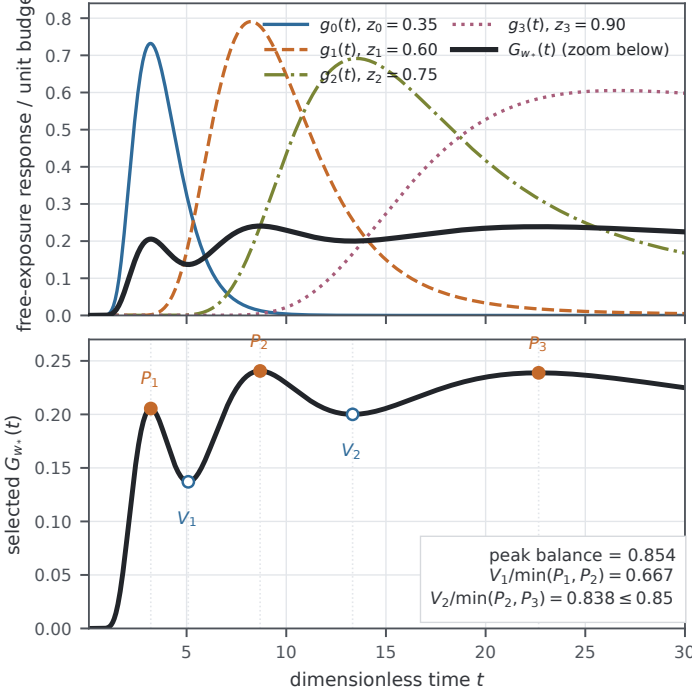
C. $B = 0$ finite-volume bridge and discrete history

A separate result-informed $B = 0$ bridge uses the same centres but broader patch and initial half-widths 0.04 and 0.02. Its exact unbounded-kernel cusp is at $t = 13.30724696$, with scaled fourth derivative -44.6816 and unfolding singular-value ratio 0.2649. A mesh-robustness rule, distinct from the exact-only selection, chose inward step $s = 0.13$ and weights $(0.28000, 0.27737, 0.08572, 0.35692)$. On the one fixed reflecting box of Eq. (32), odd cubic meshes $N = 65, 97, 129, 193$ all retain five alternating sign-changing roots; cusp-time errors decrease

(a) Four narrow catalyst slabs and the selected conserved allocation



(b) Free-exposure channels and selected mixture



Result-informed $B=0$ direct-continuum calculation: relative-shape gate PASS; no event-mass claim; continuum_verified = finite-B = project gate = false.

(c) Frozen inward-step scan

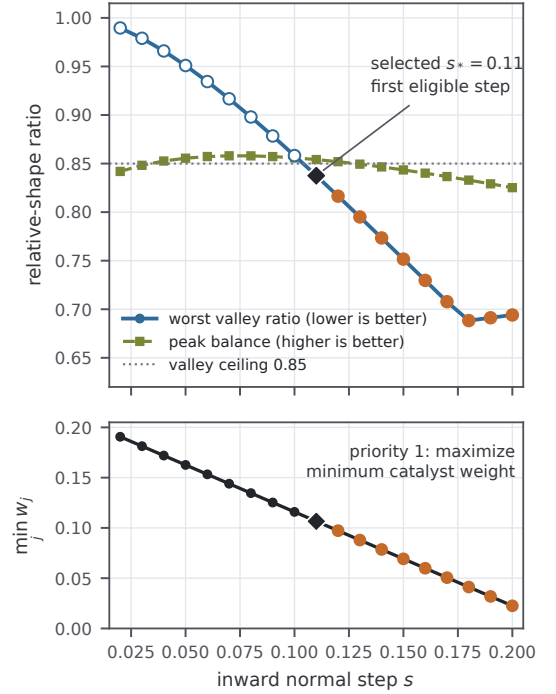


FIG. 1. Exact free-exposure design in the physical- $d = 2$ slab quotient. (a) Four normalized longitudinal catalyst slabs under one conserved installed budget. (b) The four exact continuum-kernel clocks and their selected G mixture versus dimensionless t , with three retained maxima and two intervening minima. (c) The frozen inward- step relative-shape scan and its maximum-minimum-weight selection rule. This result-informed $B = 0$ confirmation is quadrature-converged but is not an interval-global root certificate, a positive-event-mass statement, or a finite- B killed-Doi computation.

as 0.5092, 0.2706, 0.1519, 0.0666, and maximum root-time errors as 0.5337, 0.2643, 0.1562, 0.0669. The two finest meshes also pass the declared relative peak/valley gates. This is **[result-informed $B = 0$ numerical bridge]**, not a preregistered discovery, interval certificate, positive- B Doi result, unbounded-box limit, physical- $d = 3$ calculation, or project gate. It does not replace the frozen half-width-0.008 chain above.

D. A fixed positive-budget multimodal point

The exact $B \downarrow 0$ calculation was used to choose the broad support family and the allocation (0.28000000, 0.27736690, 0.08571723, 0.35691587). With

these physical inputs held fixed and no positive-budget refit, we evaluated the killed-Doi law at $B = 0.01$ in the same reflected box on the odd meshes $N = 113$ and $N = 129$. Both calculations retain the max-min-max-min-max topology among five retained roots on the saved root screen $t \leq 35$, and the paired root times differ by at most 0.03408. Peak ratios lie in $[0.83339-0.83914]$; the early and late valley ratios lie in $[0.76468-0.78239]$ and $[0.84375-0.84673]$. Partitioning reaction probability at the two valleys gives the three basin-mass ranges $[0.00521143-0.00522784]$, $[0.01659738-0.01662829]$, and $[0.14837901-0.14848157]$ through $t = 100$. Final survival lies in $[0.82969321-0.82978127]$; the largest basin-mass and survival differences are 1.03×10^{-4} and 8.81×10^{-5} .

For one unchanged positive-budget control, this cal-

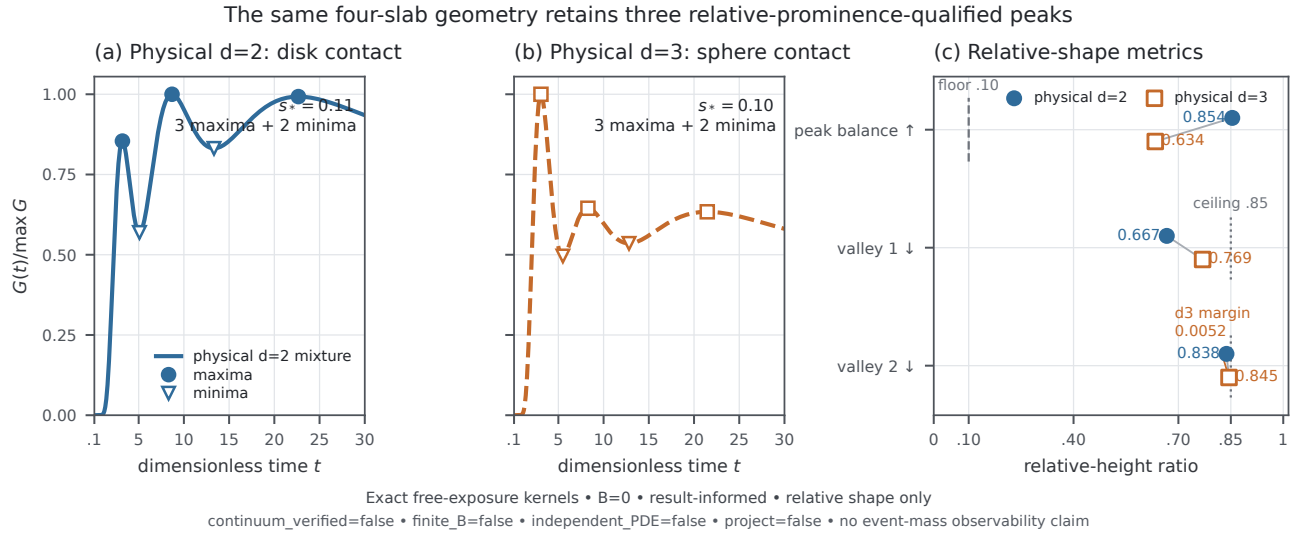


FIG. 2. Dimension comparison for the same four-slab centres, transport, contact radius, and frozen selection protocol. (a) The physical- $d = 2$ disk-contact and (b) physical- $d = 3$ sphere-contact selected dimensionless-time $B = 0$ free-exposure curves, each normalized only for shape comparison. Their selected allocations differ because each dimension applies the frozen inward-step rule to its own exact kernel. (c) Relative peak-balance and valley ratios against the declared shape thresholds. Both dimensions retain three qualified maxima, but the $d = 3$ second-valley margin is only 0.0052. This result-informed comparison does not establish positive event mass, finite- B persistence, an independent killed-PDE solve, or a project gate.

culation establishes at least three event-mass-qualified modes in the valley-partitioned sense through $t = 100$, based on five retained roots on the saved screen $t \leq 35$; it does not exclude additional extrema after $t = 35$. The first basin clears the numerical floor by only 4.23%, and the worst valley ratio is 0.84673 against the 0.85 ceiling. These tight margins make parity/alignment, box, and independent-method continuation scientifically necessary; two odd meshes in one box cannot establish an unbounded or continuum limit, and one control cannot establish a cusp.

An earlier three-slab, 207 025-state free-exposure diagnostic had a well-conditioned local cusp but no sampled trimodal control: the complete 0.01 simplex contained 4696 one-mode and 455 two-mode controls. Its shallow extra pair motivated the four-channel construction; it is retained as audit history rather than promoted as physical trimodality evidence.

The implemented cell-centred solver uses Scharfetter–Gummel longitudinal fluxes, periodic transverse diffusion, cell-integrated smooth bumps, and adaptive circle-rectangle contact fractions checked cell by cell against an independent order-128 Gauss–Legendre construction. Its current $25^3 = 15\,625$ -state smoke passes 42 stage-specific gates (38 reusable foundation gates and four solver gates), including:

- full-budget relative error 1.85×10^{-16} , maximum patch-integral error 4.44×10^{-16} , and endpoint-budget error 3.70×10^{-16} ;
- maximum per-cell contact-reference error $2.89 \times$

10^{-15} and centroid magnitude 5.96×10^{-17} ;

- exactly zero discretized initial contact mass;
- free row-sum and killed mass-balance errors 2.69×10^{-15} ; and
- dense-sparse semigroup discrepancy 4.41×10^{-15} and relative finite-difference jet discrepancies no larger than 1.75×10^{-8} .

These results are exact algebraic invariants of the assembled discrete operator and an **[implementation smoke pass]** for the implementation. They are not a continuum-kernel or continuum-PDE confirmation. The subsequent G1b-informed but prospectively frozen $65 \times 65 \times 49 = 207\,025$ -state G1c discovery line propagated eleven controls on $t \in [0, 80]$ with spacing 0.25, using the generator actions $K, \mathcal{T}_\kappa K, \mathcal{T}_\kappa^2 K, \mathcal{T}_\kappa^3 K$. Every control retained exactly one sign-changing root of f_t on this screen, a non-degenerate density maximum. No near-zero extremum or adjacent-control sign bracket was found. At one control, $\theta = 0.7$, an unmatched late maximum–minimum pair of f_t triggered the predeclared manual-review guard. Both extrema remained below zero; the late maximum was -1.3839×10^{-3} and its dimensionless height was 0.51295, compared with the discovery cutoff 0.05.

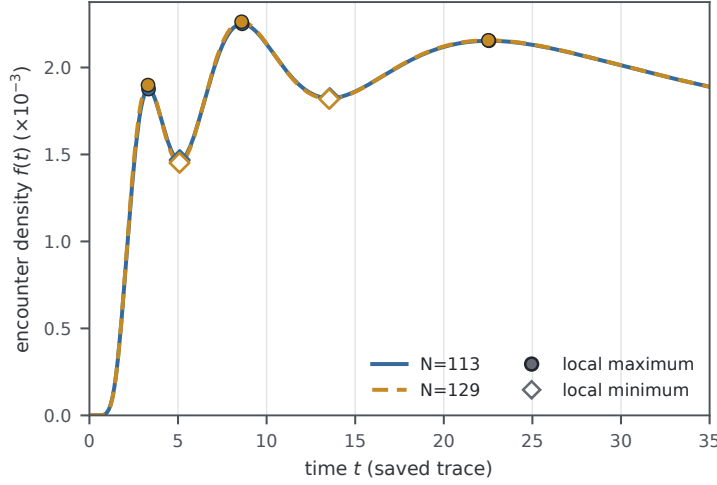
A separately frozen, explicitly post-result calculation repeated that control with time spacing 0.05. It reproduced the four f_t extrema and exactly one density maximum; the extra local maximum was -1.3837×10^{-3} . Common-time observable discrepancies from the formal run were at most 5.22×10^{-14} . This is a resolved negative

Positive-budget broad-four-slab fixed-control evidence

Pinned canonical result: five alternating roots and three qualified event basins on $N = 113, 129$.

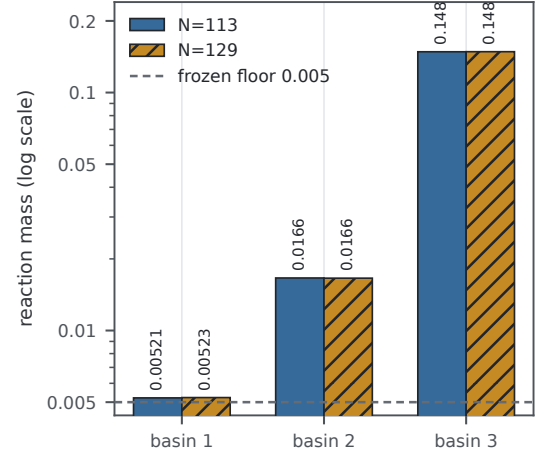
(a) Saved density traces and stationary roots

fixed reflected box; $B = 0.01$; same solver family; saved trace $t \leq 35$



(b) Event-basin reaction masses

odd meshes; basin-mass/tail gate through $t \leq 100$



Fixed box, $B = 0.01$, same solver: five retained roots on saved screen $t \leq 35$; basin-mass/tail checks through $t = 100$.

No interval-global/post-35 root exclusion, continuum/box/parity, independent-solver, physical- $d = 3$, cusp, or publication claim.

FIG. 3. Fixed-control positive-budget evidence in one broad physical- $d = 2$ family. (a) Saved density traces and the five alternating stationary roots on two odd finite-volume meshes on the saved root screen $t \leq 35$. (b) Valley-partitioned reaction masses and tail checks through $t = 100$; all six basin masses exceed the frozen 0.005 numerical floor. No interval-global or post-35 root exclusion is claimed. The $B \downarrow 0$ design fixed the allocation before these positive- B calculations. The comparison uses one reflected box and one solver family; it is not a parity, box, continuum, allocation-cusp, independent-solver, or physical- $d = 3$ result.

derivative wiggle, not a fold or a second mode. Because the original protocol also required complete adjacent-control extremum matching before declaring the line empty, the formal line action remains inconclusive rather than being retroactively relabelled.

E. A result-informed finite-budget finite-grid fold

A G1b-informed but prospectively frozen full-simplex experiment evaluated 66 triangular-lattice controls and all 165 neighboring edges at $B = 0.6$. On the frozen $t \in [0, 80]$, $\Delta t = 0.25$ screen it found 3 strict-interior interpolated seeds; post-result topology review separated 45 equal-retained-root-count matching ambiguities from eight edges with one versus three retained sign-changing roots. A deterministic largest-minimum-weight rule selected only the segment

$$\mathbf{w}(\lambda) = (0.2, 0.1\lambda, 0.8 - 0.1\lambda). \quad (40)$$

The segment was selected after G1c. A result-informed, post-G1c augmented state/sensitivity solve whose numerical choices were frozen before execution, but which was not a preregistered discovery, solves $f_t = f_{tt} = 0$

on the unchanged bounded-box 207 025-state mesh and gives

$$\begin{aligned} t_f &= 10.5022583145, & \lambda_f &= 0.6388077421, \\ \mathbf{w}_f &= (0.20000000, 0.06388077, \\ & \quad 0.73611923), \\ f(t_f) &= 0.0256240551. \end{aligned} \quad (41)$$

The scaled fold jet is

$$\begin{aligned} t f_t / f &= 2.84 \times 10^{-15}, & t^2 f_{tt} / f &= 2.39 \times 10^{-13}, \\ t^3 f_{ttt} / f &= -7.9231, & t f_{t\lambda} / f &= -0.32729, \\ t^2 f_{tt\lambda} / f &= -0.61492. \end{aligned} \quad (42)$$

and the corresponding dimensionless Jacobian determinant is -2.59316 . On the frozen G1d $t \in [3, 18]$, $\Delta t = 0.02$ sign-changing-root screen, $\lambda_f - 0.02$ retains alternating maximum-minimum-maximum roots, whereas $\lambda_f + 0.02$ retains one maximum. This is not a global root census and does not exclude even-multiplicity or subgrid pairs. Exact augmented sensitivities agree with decreasing-step centred control differences within the frozen tolerance. This is **[numerically verified—discrete diagnostic]** for one finite-budget finite-grid

fold. It is not a continuum extrapolation, a complete phase map, a cusp, or a trimodality result.

The next confirmation must converge the entire jet in Eq. (10), not only the fold coordinates. Odd/even meshes, box enlargement, contact quadrature, and an independent off-lattice or radiation-boundary method must agree. A finite- B four-slab cusp requires every entry of Eq. (12), projected rank two, and five alternating simple roots with the declared prominence and event-mass floors.

VII. CURRENT GATE LEDGER

The scoped analytical existence gate is met, and the $B = 0$ continuum-kernel cusp plus the bounded-grid finite- B fold justify continuing the PRR route. Submission status and stronger headline wording remain withheld until the same broad family has a finite- B allocation cusp and both folds, the complete fold/cusp jets and fixed-control event-law features persist under parity/alignment/box refinement, and an independent method preserves the positive-budget event-law features. A two-/three-dimensional headline also requires separate positive-budget physical- $d = 3$ evidence. The project is redirected if parity sequences disagree, nondegeneracy vanishes, independent methods change the topology, or secondary modes fall below the declared prominence and event-mass floors.

VIII. DISCUSSION

The organizing object is the zero set of f_t on a conserved-resource control manifold, not the number of catalyst patches or a particular lattice. For every fixed finite integer $d \geq 2$ and each prescribed fixed finite m , the direct theorem constructs a d - and m -dependent longitudinal-slab OU quotient whose Doi density has at least m nondegenerate maxima after first choosing a sufficiently small joint noise/initial-variance/slab-width parameter ϵ and then a sufficiently small positive budget. It does not fix one transport and geometry across d or m , give a uniform-in- d or $d \rightarrow \infty$ result, determine the global number of modes, or provide an event-mass floor; it deliberately keeps the relative deterministic path inside contact near the designed peaks.

The result-informed four-slab calculations exhibit three relative-prominence-qualified maxima and a cusp on the declared floating-point $B = 0$ screens for both the physical- $d = 2$ disk and physical- $d = 3$ sphere kernels. For the separate broad physical- $d = 2$ family, one unchanged allocation at $B = 0.01$ retains three event-mass-qualified modes on two odd fixed-box meshes in a valley-partitioned check through $t = 100$, using five retained roots on the saved screen $t \leq 35$; no post-35 or interval-global root exclusion is available. Its first-basin and valley margins are tight and the two meshes use the

same solver. The smaller exact-kernel $d = 3$ peak ratio and its 0.0052 worst-valley margin identify the three-dimensional design as the more fragile positive-budget test. The post-G1c G1d calculation shows an interior fold at $B = 0.6$ on one bounded reflected finite-volume model. Neither numerical result closes the other's missing limit.

The decisive remaining physics is whether fixed-budget allocation coordinates produce a same-family cusp and both folds, whether the resulting jets and representative event laws survive parity/alignment/box continuation, and whether an independent off-lattice killed process preserves the unchanged-control features within quantified uncertainty. Positive-budget $d = 3$ is a separate later branch rather than evidence borrowed from the exact $B = 0$ sphere kernel. Further reduced-clock scans do not address these gates.

ACKNOWLEDGMENTS

Funding and computational-resource statements are pending explicit author confirmation. Author order, companion-manuscript disclosure, data/code availability, and archival identifiers remain hard release gates.

Appendix A: Tail and localization estimates for the direct theorem

This appendix records the estimates behind the direct physical theorem. The midpoint OU process in Eq. (24) has variance $\epsilon^2 s^2(t)$, where

$$s^2(t) = s_0^2 e^{-2\gamma t} + \frac{D_0}{2\gamma} (1 - e^{-2\gamma t}). \quad (\text{A1})$$

Its invariant variance is $\epsilon^2 D_0/(2\gamma)$. Direct comparison of Gaussian exponents shows that the declared Gaussian initial density belongs to $L^2(\pi_\epsilon^{-1})$ precisely when its variance is less than twice the invariant variance, giving $s_0^2 < D_0/\gamma$. For the longitudinal relative OU process, the variance coefficient is

$$u_0^2 e^{-2\gamma t} + \frac{2D_0}{\gamma} (1 - e^{-2\gamma t}), \quad (\text{A2})$$

and the analogous condition is $u_0^2 < 4D_0/\gamma$. A wrapped Gaussian is square integrable against the uniform torus measure for every fixed positive ϵ . These observations justify the weighted-space corollary in Eq. (17); none is uniform as $\epsilon \downarrow 0$.

Let I_* be the union of the target-time neighborhoods and suppose the deterministic relative mean is at least distance $\eta > 0$ inside the contact ball there. The relative transition law is a wrapped Gaussian with covariance $\epsilon^2 \Sigma_R(t)$, where Σ_R and its time derivatives are bounded and Σ_R is uniformly positive definite on I_* . In the product geodesic distance on $\mathbb{R} \times \mathbb{T}_W^{d-1}$, the reverse triangle inequality places the complement of the

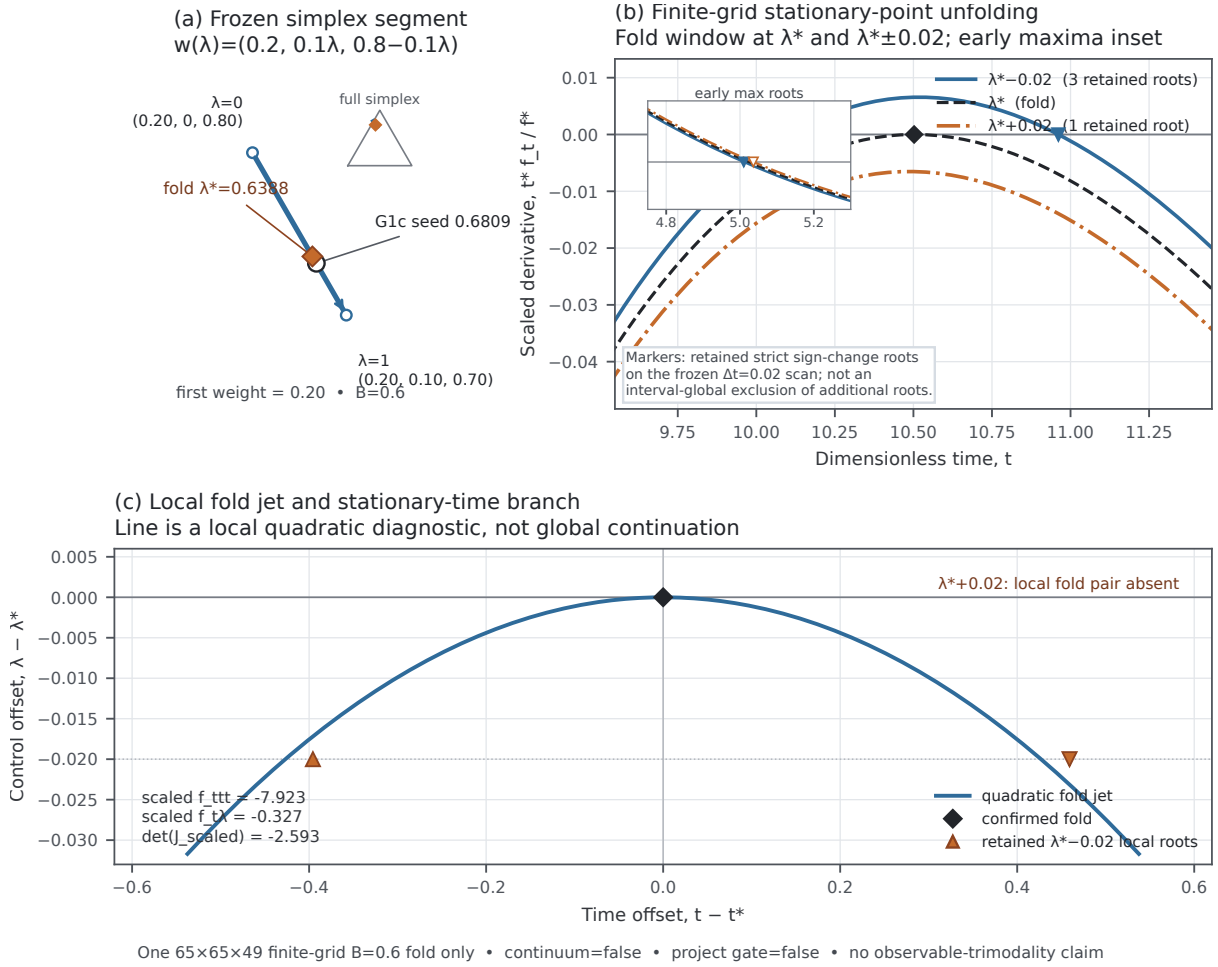


FIG. 4. One bounded reflected $65 \times 65 \times 49$ finite-volume Doi model at $B = 0.6$ only. The result-informed segment was selected after G1c; its numerical choices were frozen before the G1d solve, but it was not a preregistered discovery. The side panels report retained strict sign-changing roots on the frozen $t \in [3, 18]$, $\Delta t = 0.02$ screen. The figure is not an interval-global root census, a continuum extrapolation, a trimodality result, or an independent-solver check.

contact ball at distance at least η from the mean. Differentiating each Gaussian lattice image r times in t gives that image times a polynomial and a fixed power of ϵ^{-1} . Fixed-dimensional lattice shells contain only polynomially many images while their Gaussian distances grow linearly with shell index. Integrating the resulting summable majorant outside the contact ball yields

$$\sup_{t \in I_*} |\partial_t^r \{c_{d,\epsilon}(t) - 1\}| \leq C_{r,d} \epsilon^{-N_{r,d}} \exp(-q_d/\epsilon^2). \quad (\text{A3})$$

The condition $a < W/2$ keeps the embedded contact-ball boundary away from the torus cut locus; no single chart for its complement is assumed. Termwise time differentiation is justified by the same uniform shell majorant on the compact positive-time interval. All constants may depend on the fixed finite d .

Convoluting the normalized Gaussian slab in Eq. (26) with the midpoint transition density gives Eq. (27) exactly. For $t = t_j + \epsilon y$, Taylor expansion on fixed $|y| \leq L$

gives

$$\frac{c_j - \mu(t_j + \epsilon y)}{\epsilon} = -\mu'(t_j)y + O(\epsilon) \quad (\text{A4})$$

in C^2 . Smooth composition with the Gaussian formula and Eq. (A3) proves the C^2 convergence in Eq. (28). If $i \neq j$, strict monotonicity implies $|c_i - \mu(t)| \geq |c_i - c_j|/2$ on the shrinking j th interval for all small ϵ ; time differentiation adds only powers of ϵ^{-1} and proves Eq. (29). Exponential cross-channel decay therefore dominates the own-channel slope and curvature powers uniformly over every compact simplex-interior weight set.

Finally, after fixing ϵ , all Gaussian catalysts are bounded. The compact-time C^2 instance of the unbounded weighted-space corollary following Eq. (16) preserves the two endpoint-slope signs and strict negative curvature on every certified local interval. This supplies one unique nondegenerate positive- B Doi maximum per target interval. Opposite derivative signs on consecutive

TABLE I. Current evidence. Each status refers only to the scope in the middle column.

Item	Evidence scope	Status
reduced GIG screens	inherited $m = 2, \dots, 6$ roots/cusp; no physical realization	PASS REDUCED ONLY
multi-rate jet algebra	fixed-support, budget-tangent local theorem	PROVED
fixed finite $d \geq 2$ and finite m	admissible monotone/contact-interior targets; d -, m -, and ϵ -dependent slabs; first ϵ , then B ; at least m ; no uniform-in- d claim	PROVED (SCOPED)
G1a operator smoke	one bounded reflected box; discrete foundations and reference checks	SMOKE ONLY
narrow four-slab free exposure	result-informed exact unbounded $d = 2$ kernels; $B = 0$; floating screen and relative shape only	PASS: RESULT-INFORMED $B = 0$ SHAPE
broad four-slab mesh bridge	result-informed $B = 0$ exact kernels plus four meshes on one fixed reflected box	PASS: $B = 0$ FIXED-BOX BRIDGE
finite- B fold	post-G1c result-informed mixed jet on one bounded 207 025-state grid	PASS: ONE BOUNDED GRID
fold convergence	odd/even mesh, box enlargement, independent solver	NOT RUN
positive- B	unchanged $B = 0.01$ broad allocation; five retained roots on the saved screen $t \leq 35$; valley-partitioned basin mass/tail checks	PASS: FIXED
four-slab point	through $t = 100$; no post-35 exclusion; fixed-box odd meshes $N = 113, 129$	CONTROL/TWO MESHES ONLY
positive- B four-slab cusp	allocation cusp, both folds, and representative modal regions in the same broad family	NOT RUN
weak- B persistence	bounded and unbounded-weighted compact-time mixed jets; application requires explicit margins	PROVED; APPLICATION NEEDS MARGINS
global GIG realization	optional physical mixed-jet asymptotic	OPEN
physical $d = 3$ sphere kernel	result-informed exact $B = 0$ cusp and relative shape; no killed-PDE event mass	PASS: $B = 0$ SHAPE
positive- B $d = 3$ transition	event mass, mesh/box convergence, independent solver	NOT RUN

gap endpoints force at least one interior local minimum,

but neither nondegeneracy of those minima nor absence of additional extrema is asserted.

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